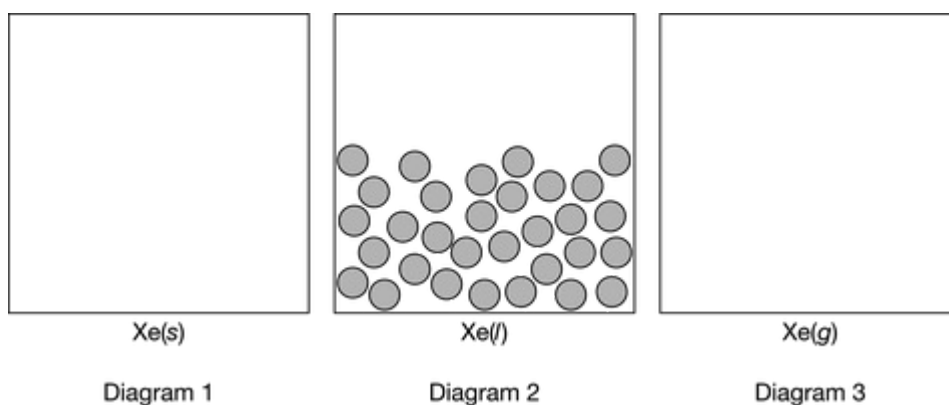


Unit 3 Progress Check: FRQ

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to **Xe**.



(a) Diagram 2 shown above represents a particle-level view of liquid **Xe** atoms.

- (i) In the box in diagram 1, draw a particle-level view that represents a vessel that contains only solid **Xe**.



Please respond on separate paper, following directions from your teacher.

- (ii) In the box in diagram 2, draw a particle-level view that represents a vessel containing only gaseous **Xe**.



Please respond on separate paper, following directions from your teacher.

- (b) Let ***P*** and ***V*** represent the pressure and volume of the **Xe(g)** in the container in diagram 3. If a piston is used to reduce the volume of the gas to $\frac{V}{2}$ at a constant temperature, what is the new pressure in the container in terms of the original pressure, ***P***?



Please respond on separate paper, following directions from your teacher.

- (c) Describe what happens to the average speed of the **Xe** atoms in the container in diagram 3 as the



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original volume V is reduced to $\frac{V}{2}$ at a constant temperature. Explain.



Please respond on separate paper, following directions from your teacher.

Part (a)(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The drawing meets both of the criteria below:

- ☐ The circles are closer together (at least no farther apart) than they are in the middle box.
- ☐ The circles are in an ordered arrangement at the bottom of the box

Part (a)(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The drawing shows the circles much farther apart and more spread throughout than in the middle box.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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0	1
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The response indicates **2P** (or double the original **P**).

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the criteria below:

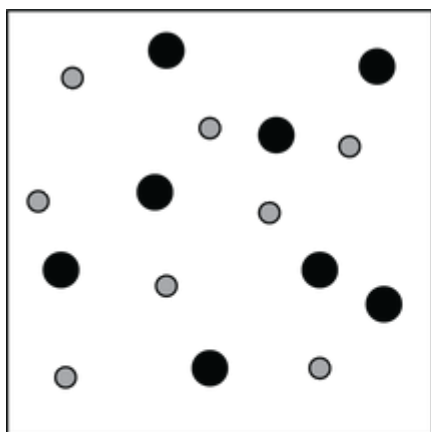
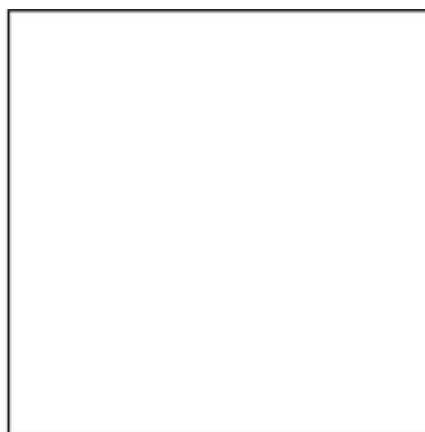
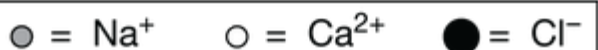
- ☐ The response indicates that the average speed stays the same (or does not change or nothing happens).
- ☐ The response indicates that the temperature is constant.

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about solutions of **NaCl** and **CaCl₂**.



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 0.20 M NaCl  0.10 M CaCl_2 

(a) Ions in a certain volume of 0.20 M NaCl(aq) are represented in the box above on the left. In the box above on the right, draw a particulate diagram that represents the ions in the same volume of $0.10\text{ M CaCl}_2\text{(aq)}$.



Please respond on separate paper, following directions from your teacher.

(b) Briefly explain how the presence of charged ions in ionic solids such as NaCl(s) helps to explain why NaCl(s) is soluble in water.



Please respond on separate paper, following directions from your teacher.

(c) What volume of 0.20 M NaCl(aq) contains 10.0 g of NaCl (molar mass 58 g/mol)?



Please respond on separate paper, following directions from your teacher.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. If the student draws two types of circles with the same shading, one small and one large, the small circles can be interpreted as Ca^{2+} ions and the large circles can be interpreted as Cl^- ions.



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0	1	2
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The drawing meets both of the criteria below:

- ☐ There are four unshaded circles that represent Ca^{2+} ions.
- ☐ There are eight shaded circles that represent Cl^- ions.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the charged ions are attracted to polar water molecules OR that the positive and negative ions are attracted to the water molecules, which have positive and negative ends.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives one of the following calculations (or an equivalent):

- ☐ $10.0 \text{ g NaCl} \times \frac{1 \text{ mol NaCl}}{58 \text{ g NaCl}} \times \frac{1000 \text{ mL}}{0.20 \text{ mol NaCl}} = 860 \text{ mL}$ (Any value from 850 mL to 862 mL is acceptable. No deduction for incorrect significant figures.)
- ☐ $10.0 \text{ g NaCl} \times \frac{1 \text{ mol NaCl}}{58 \text{ g NaCl}} = 0.17 \text{ mol NaCl}$
 $\frac{0.20 \text{ mol NaCl}}{1000 \text{ mL}} = \frac{0.17 \text{ mol NaCl}}{x \text{ mL}} \Rightarrow x = 850 \text{ mL}$



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